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AERIAL VEHICLE FLUID CONTROL SYSTEM WITH MULTI-WAY FLOW REGULATOR

Abstract

An assembly is provided that includes a regulator manifold and a regulator element. The regulator manifold includes an inlet passage, a first outlet passage and a second outlet passage. The inlet passage is configured to receive bleed gas from a core flowpath. The regulator element is within the regulator manifold and configured to pivot to a first position, a second position and an intermediate position between the first position and the second position. The regulator element is configured to fluidly couple the inlet passage with the first outlet passage in the first position. The regulator element is configured to fluidly couple the inlet passage with the second outlet passage in the second position. The regulator element is configured to fluidly decouple the inlet passage from the first outlet passage and the second outlet passage in the intermediate position.

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Background/Summary

[0001] This application is a divisional of U.S. patent application Ser. No. 17/902,261 filed Sep. 2, 2022, which is hereby incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Technical Field

[0002] This disclosure relates generally to a fluid system and, more particularly, to a fluid control system for a gas turbine engine.

2. Background Information

[0003] An aerial vehicle such as an aircraft and/or a gas turbine engine powering the aircraft may include various fluid systems for controlling aircraft and/or engine operation. The aircraft may also include a discrete flight control system for maneuvering the aircraft during flight. Various types and configurations of fluid systems and flight control systems are known in the art. While these known fluid systems and flight control systems have various benefits, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

[0004] According to an aspect of the present disclosure, an assembly is provided that includes a regulator manifold and a regulator element. The regulator manifold includes an inlet passage, a first outlet passage and a second outlet passage. The inlet passage is configured to receive bleed gas from a core flowpath of a gas turbine engine. The regulator element is within the regulator manifold and configured to pivot to a first position, a second position and an intermediate position between the first position and the second position. The regulator element is configured to fluidly couple the inlet passage with the first outlet passage in the first position. The regulator element is configured to fluidly couple the inlet passage with the second outlet passage in the second position. The regulator element is configured to fluidly decouple the inlet passage from the first outlet passage and the second outlet passage in the intermediate position.

[0005] According to another aspect of the present disclosure, an assembly is provided that includes a gas turbine engine, a regulator manifold and a regulator element. The gas turbine engine includes a compressor section, a combustor section, a turbine section and a core flowpath extending through the compressor section, the combustor section and the turbine section. The regulator manifold includes an inlet passage, a first outlet passage and a second outlet passage. The inlet passage is configured to receive bleed gas from the core flowpath. The regulator element is within the regulator manifold and configured to pivot to a first position, a second position and an intermediate position between the first position and the second position. The regulator element is configured to fluidly couple the inlet passage with the first outlet passage in the first position. The regulator element is configured to fluidly couple the inlet passage with the second outlet passage in the second position. The regulator element is configured to fluidly decouple the inlet passage from the first outlet passage and the second outlet passage in the intermediate position.

[0006] According to another aspect of the present disclosure, another assembly is provided that includes a component and a fluid circuit. The component is outside of a gas turbine engine. The fluid circuit is configured to bleed gas from a core flowpath of the gas turbine engine to provide bleed gas. The fluid circuit is configured to direct the bleed gas to the component during a first

mode. The fluid circuit is configured to direct the bleed gas back to the core flowpath and/or into a plenum outside of the gas turbine engine during a second mode.

[0007] According to still another aspect of the present disclosure, another assembly is provided that includes a regulator manifold and a regulator element. The regulator manifold includes an inlet passage, a first outlet passage, a second outlet passage and a third outlet passage. The regulator element is within the regulator manifold. The regulator element is configured to pivot to a first position, a second position and an intermediate position between the first position and the second position. The regulator element is configured to fluidly couple the inlet passage with the first outlet passage and fluidly decouple the inlet passage from the second outlet passage and the third outlet passage in the first position. The regulator element is configured to fluidly couple the inlet passage with the second outlet passage and fluidly decouple the inlet passage from the first outlet passage and the third outlet passage in the second position. The regulator element is configured to fluidly couple the inlet passage with the third outlet passage and fluidly decouple the inlet passage from the first outlet passage and the second outlet passage in the intermediate position.

[0008] The regulator element may be configured to flow gas to a first side of the regulator element from the inlet passage to the first outlet passage when the regulator element is in the first position. The regulator element may be configured to flow the gas to a second side of the regulator element from the inlet passage to the second outlet passage when the regulator element is in the second position. The regulator element may be configured to flow the gas through an interior of the regulator element from the inlet passage to the third outlet passage when the regulator element is in the intermediate position.

[0009] The component may be configured as or otherwise include an airfoil configured as a control surface.

[0010] The fluid circuit may be configured to direct the bleed gas to a first portion of the component during the first mode. The fluid circuit may be configured to direct the bleed gas to a second portion of the component during a third mode.

[0011] The fluid circuit may include a movable regulator element. The fluid circuit may be configured to flow the bleed gas to a side of the movable regulator element during the first mode. The fluid circuit may be configured to flow the bleed gas through an internal volume of the movable regulator element during the second mode.

[0012] The regulator element may be configured to fluidly decouple the inlet passage from the second outlet passage in the first position. The regulator element may be configured to fluidly decouple the inlet passage from the first outlet passage in the second position.

[0013] The assembly may also include an airfoil extending laterally between a first surface and a second surface. The airfoil may include a first aperture in the first surface and a second aperture in the second surface. The first outlet passage may be fluidly coupled with the first aperture. The second outlet passage may be fluidly coupled with the second aperture.

[0014] The assembly may also include an airframe including a body and the airfoil projecting out from the body. The gas turbine engine may be mounted to the airframe.

[0015] The regulator manifold may also include a third outlet passage. The regulator element may be configured to fluidly couple the inlet passage to the third outlet passage in the intermediate position.

[0016] The assembly may also include an airfoil including a first surface and a first aperture in the first surface. The first outlet passage may be fluidly coupled with the first aperture.

[0017] The airfoil may also include a second surface and a second aperture in the second surface. The second outlet passage may be fluidly coupled with the second aperture.

[0018] The third outlet passage may be configured to fluidly couple the inlet passage with the core flowpath when the regulator element is in the intermediate position.

[0019] The third outlet passage may be configured to fluidly couple the inlet passage with a plenum outside of the gas turbine engine when the regulator element is in the intermediate position.

[0020] The regulator element may include an internal volume configured to fluidly couple the inlet passage to the third outlet passage when the regulator element is in the intermediate position.

[0021] The regulator element may include a first channel and a second channel. The first channel may fluidly couple the inlet passage with the first outlet passage when the regulator element is in the first position. The second channel may fluidly couple the inlet passage with the second outlet passage when the regulator element is in the second position.

[0022] The gas turbine engine may include a rotating structure and a stationary structure. The rotating structure may include a compressor rotor within a compressor section of the gas turbine engine and a turbine rotor within a turbine section of the gas turbine engine. The stationary structure may at least partially house the rotating structure. The regulator manifold may be formed integral with the stationary structure.

[0023] The stationary structure may house the turbine rotor.

[0024] The stationary structure may include a combustor within a combustor section of the gas turbine engine.

[0025] The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

[0026] The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a partial side sectional illustration of an assembly for an aerial vehicle.

[0028] FIG. 2 is a partial side sectional illustration of the vehicle assembly at a central portion of a gas turbine engine.

[0029] FIG. 3 is a perspective illustration of a portion of a vehicle body.

[0030] FIG. 4 is a side illustration of a portion of the vehicle body at an airfoil forming a control surface for the aerial vehicle.

[0031] FIG. 5 is a sectional illustration of a portion of the vehicle body at the airfoil.

[0032] FIG. 6 is a cross-sectional illustration of the airfoil.

[0033] FIGS. 7A and 7B are cross-sectional illustrations of an intake duct at an inlet with various inlet aperture arrangements.

[0034] FIG. 8 is a schematic illustration of a fluid control system fluidly coupled with a core flowpath of the gas turbine engine.

[0035] FIGS. 9A and 9B are sectional illustrations of a portion of a fluid circuit for the fluid control system with a flow regulator element in various side positions.

[0036] FIGS. 10A and 10B are cross-sectional illustrations of the airfoil during various modes of operation.

[0037] FIG. 11 is a sectional illustration of a portion of the fluid circuit with the flow regulator element in an intermediate position.

[0038] FIGS. 12A-C are sectional illustrations of a portion of the fluid circuit for the fluid control system with another flow regulator element in various positions.

[0039] FIGS. 13A-C are sectional illustrations of a portion of the fluid circuit for the fluid control system with still another flow regulator element in various positions.

[0040] FIG. 14 is a schematic illustration of the fluid control system fluidly coupled with the core flowpath and a plenum outside of the gas turbine engine.

[0041] FIG. 15 is a schematic illustration of the fluid control system fluidly coupled with the core flowpath at multiple locations.

DETAILED DESCRIPTION

[0042] FIG. 1 illustrates an assembly 20 for an aerial vehicle such as, but not limited to, an unmanned aerial vehicle (UAV), a drone or any other manned or unmanned aircraft or self-propelled projectile. The vehicle assembly 20 of FIG. 1 includes a gas turbine engine 22, a vehicle airframe 24 and a fluid control system 26 that integrates the gas turbine engine 22 with the vehicle airframe 24 to provide, for example, vehicle flight control.

[0043] The gas turbine engine 22 of FIG. 1 is configured as a single spool, radial-flow turbojet gas turbine engine. This gas turbine engine 22 is configured for propelling the aerial vehicle. The present disclosure, however, is not limited to such an exemplary turbojet gas turbine engine configuration nor to a vehicle propulsion system application. For example, the gas turbine engine 22 may alternatively be configured as an auxiliary power unit (APU) for the aerial vehicle.

[0044] The gas turbine engine 22 of FIG. 1 extends axially along an axial centerline 28 between a forward, upstream airflow inlet 30 into the gas turbine engine 22 and an aft, downstream exhaust 32 out of the gas turbine engine 22. This axial centerline 28 may be a centerline axis of the gas turbine engine 22 and the vehicle body 110. The axial centerline 28 may also or alternatively be a rotational axis for one or more components within the gas turbine engine 22.

[0045] The gas turbine engine 22 includes a core flowpath 34, an inlet section 36, a (e.g., radial flow) compressor section 37, a (e.g., reverse flow) combustor section 38, a (e.g., radial flow) turbine section 39 and an exhaust section 40. The gas turbine engine 22 also includes a static engine structure 42 housing and/or forming one or more or all of the engine section 36-40.

[0046] The core flowpath 34 extends within the gas turbine engine 22 between the engine inlet 30 and the engine exhaust 32. More particularly, the core flowpath 34 extends sequentially through the inlet section 36, the compressor section 37, the combustor section 38, the turbine section 39 and the exhaust section 40 from the engine inlet 30 to the engine exhaust 32.

[0047] The compressor section 37 includes a bladed compressor rotor 44. The turbine section 39 includes a bladed turbine rotor 46. Each of these engine rotors 44, 46 includes a plurality of rotor blades arranged circumferentially around and connected to at least one respective rotor disk and/or hub. The rotor blades, for example, may be formed integral with or mechanically fastened, welded, brazed, adhered and/or otherwise attached to the respective rotor disk(s) and/or hub(s).

[0048] The compressor rotor 44 may be configured as a radial flow compressor rotor. The turbine rotor 46 may be configured as a radial flow turbine rotor. The compressor rotor 44 is connected to the turbine rotor 46 through an engine shaft 48. This engine shaft 48 is rotatably supported by the static engine structure 42 through a plurality of bearings 50A and 50B (generally referred to as 50); e.g., rolling element bearings, journal bearings, etc.

[0049] The combustor section 38 includes an annular combustor 52 with an annular combustion chamber 54. The combustor 52 of FIG. 1 is configured as a reverse flow combustor. Inlets ports 56/flow tubes into the combustion chamber 54, for example, may be arranged at (e.g., on, adjacent or proximate) and/or towards an aft bulkhead wall 58 of the combustor 52. An outlet from the combustor 52 may be arranged axially aft of an inlet to the turbine section 39. The combustor 52 may also be arranged radially outboard of and/or axially overlap at least a portion (e.g., aft portion) of the turbine section 39. With this arrangement, the core flowpath 34 of FIG. 1 reverses direction (e.g., from a forward-to-aft direction to an aft-to-forward direction) a first time as the core flowpath 34 extends from a diffuser plenum 60 surrounding the combustor 52 into the combustion chamber 54. The core flowpath 34 of FIG. 1 then reverses its direction (e.g., from the aft-to-forward direction to the forward-to-aft direction) a second time as the core flowpath 34 extends from the combustion chamber 54 into the turbine section 39.

[0050] During operation, air enters the gas turbine engine 22 through the inlet section 36 and its engine inlet 30. The inlet section 36 directs this air from the engine inlet 30 into the core flowpath 34 and the compressor section 37. The air within the core flowpath 34 may be referred to as core air. This core air is compressed by the compressor rotor 44 and directed through a diffuser 62 and its plenum 60 into the combustion chamber 54. Fuel is injected and mixed with the compressed

core air to provide a fuel-air mixture. This fuel-air mixture is ignited within the combustion chamber **54**, and combustion products thereof flow through the turbine section **39** and cause the turbine rotor **46** to rotate. The rotation of the turbine rotor **46** drives rotation of the compressor rotor **44** and, thus, compression of the air received from the engine inlet **30**. The exhaust section **40** receives the combustion products from the turbine section **39**. The exhaust section **40** directs the received combustion products out of the gas turbine engine **22** to provide forward engine thrust. [0051] The static engine structure **42** of FIG. **1** includes one or more stationary structures **64** and **66**. These stationary structures **64** and **66** are configured as discrete sub-structures (e.g., axial sections) of the static engine structure **42**. The forward, upstream stationary structure **64** and/or the aft, downstream stationary structure **66** may each be formed as (or part of) a monolithic body **68**, **70** (e.g., a monolithic vehicle component), respectively. Herein, the term “monolithic” may describe a component of the vehicle which is formed as a single unitary body. Each stationary structure **64**, **66**, for example, may be additively manufactured, cast, machined and/or otherwise formed as an integral, unitary body. By contrast, a non-monolithic body may include parts that are discretely formed from one another, where those parts are subsequently mechanically fastened and/or otherwise attached to one another.

[0052] The upstream stationary structure **64** may form a section of the static engine structure **42** along any one or more or all of the engine sections **36** and **37**. The upstream stationary structure **64** of FIG. **2**, for example, includes an inlet nozzle **72** and at least a forward, upstream section **74** (or an entirety) of a compressor case **76**. The upstream stationary structure **64** may also include an internal support structure for at least one of the bearings **50**; e.g., the forward bearing **50A**.

[0053] The inlet nozzle **72** is disposed within the inlet section **36**, for example, at the engine inlet **30**. This inlet nozzle **72** may be configured to condition the core air entering the compressor section **37**. The inlet nozzle **72** of FIG. **2**, for example, includes one or more inlet guide vanes **78** configured to impart swirl to the core air. These inlet guide vanes **78** are arranged circumferentially about the axial centerline **28** in an annular array; e.g., an inlet nozzle vane array. Each of the inlet guide vanes **78** extends radially across the core flowpath **34** between and is connected to an inner platform **80** of the inlet nozzle **72** and an outer platform **82** of the inlet nozzle **72**.

[0054] The compressor case **76** houses the compressor section **37** and its compressor rotor **44**. The compressor case **76**, for example, extends circumferentially about (e.g., circumscribes) and axially overlaps the compressor rotor **44**. The compressor case **76** may thereby form an outer peripheral boundary of the core flowpath **34** within the compressor section **37**.

[0055] The downstream stationary structure **66** may form a section of the static engine structure **42** along any one or more or all of the engine sections **38-40**; see also FIG. **1**. The downstream stationary structure **66** of FIG. **2**, for example, includes a diffuser nozzle **84**, an outer diffuser (e.g., plenum) case **86**, the combustor **52**, a turbine nozzle **88**, a turbine case **90** and an exhaust case **92**; see also FIG. **1**. The downstream stationary structure **66** may also include an internal support structure for at least one of the bearings **50**; e.g., the aft bearing **50B**.

[0056] The diffuser nozzle **84** is disposed within the diffuser **62** upstream of the plenum **60**. This diffuser nozzle **84** may be configured to condition the core air leaving the compressor section **37** and entering the plenum **60**. The diffuser nozzle **84** of FIG. **2**, for example, includes one or more diffuser guide vanes **94** configured to impart swirl to the core air. These diffuser guide vanes **94** are arranged circumferentially about the axial centerline **28** in an annular array; e.g., a diffuser nozzle vane array. Each of the diffuser guide vanes **94** extends radially across the core flowpath **34** between and is connected to an inner platform **96** of the diffuser nozzle **84** and an outer platform **98** of the diffuser nozzle **84**.

[0057] The diffuser case **86** houses the combustor **52** within the combustor section **38**. The diffuser case **86**, for example, extends circumferentially about (e.g., circumscribes), axially overlaps and is spaced radially outward from the combustor **52**. The diffuser case **86** may thereby form an outer peripheral boundary of the core flowpath **34** and the plenum **60** along the combustor **52**. The outer

platform **98** of the diffuser nozzle **84** may be formed as a forward, upstream extension of the diffuser case **86**.

[0058] The combustor **52** of FIG. **2** includes an outer combustor wall **100**, an inner combustor wall **102** and the bulkhead wall **58**. The outer combustor wall **100** extends axially between and may be connected to an outer platform **104** of the turbine nozzle **88** and the bulkhead wall **58**. The outer combustor wall **100** extends circumferentially about (e.g., circumscribes), axially overlaps and is spaced radially outward from the inner combustor wall **102**. The inner combustor wall **102** extends axially between an inner platform **106** of the turbine nozzle **88** and the bulkhead wall **58**, and the inner combustor wall **102** may be connected to the bulkhead wall **58**. The inner combustor wall **102** extends circumferentially about (e.g., circumscribes), axially overlaps and is spaced radially outward from the turbine case **90**. The bulkhead wall **58** extends radially between the outer combustor wall **100** and the inner combustor wall **102**, at aft ends of the combustor walls **100** and **102**. With this arrangement, the combustor walls **58**, **100** and **102** collectively form peripheral boundaries of the combustion chamber **54** within the combustor **52**.

[0059] The turbine nozzle **88** is disposed within the turbine section **39**, for example, at an intersection between the combustor section **38** and the turbine section **39**. This turbine nozzle **88** may be configured to condition the combustion products leaving the combustor section **38** and its combustion chamber **54** and entering the turbine section **39**. The turbine nozzle **88** of FIG. **2**, for example, includes one or more turbine guide vanes **108** configured to impart swirl to the combustion products. These turbine guide vanes **108** are arranged circumferentially about the axial centerline **28** in an annular array; e.g., a turbine nozzle vane array. Each of the turbine guide vanes **108** extends radially across the core flowpath **34** between and is connected to the inner platform **106** of the turbine nozzle **88** and the outer platform **104** of the turbine nozzle **88**.

[0060] The turbine case **90** houses at least a portion of the turbine section **39** and its turbine rotor **46**. The turbine case **90**, for example, extends circumferentially about (e.g., circumscribes) and axially overlaps at least a portion (e.g., an aft, downstream portion) of the turbine rotor **46**. The turbine case **90** may thereby form an outer peripheral boundary of the core flowpath **34** within at least a portion (e.g., an aft, downstream portion) of the compressor section **37**.

[0061] Referring to FIG. **1**, the exhaust case **92** extends axially from one or more of the diffuser case **86** and/or the turbine case **90** to the engine exhaust **32**. The exhaust case **92** may also be connected to the diffuser case **86** and/or the turbine case **90** at aft ends thereof. The exhaust case **92** extends circumferentially about (e.g., completely around) the axial centerline **28**. The exhaust case **92** may thereby form an outer peripheral boundary of the core flowpath **34** within the exhaust section **40** to the engine exhaust **32**.

[0062] An aft, downstream end of the upstream stationary structure **64** is mated with and connected to a forward, upstream end of the downstream stationary structure **66**. The upstream stationary structure **64** of FIG. **1**, for example, is attached to the downstream stationary structure **66** by at least one mechanical joint; e.g., a bolted flange connection. However, in other embodiments, the upstream stationary structure **64** may also or alternatively be attached to the downstream stationary structure **66** by at least one bond joint; e.g., a braze connection, a welded connection, etc.

[0063] The gas turbine engine **22** is arranged with and mounted to the vehicle airframe **24**. The vehicle airframe **24** of FIG. **1**, for example, includes a vehicle body **110** and one or more vehicle airfoils **112**. The gas turbine engine **22** may be (e.g., completely) housed within the vehicle airframe **24** and its vehicle body **110**. The vehicle body **110** forms an outer (e.g., exterior) aerodynamic surface **114** for the aerial vehicle; see also FIG. **3**. The vehicle body **110**, for example, may be configured as a nacelle for the gas turbine engine **22**. The vehicle body **110** may also or alternatively be configured as a fuselage and/or another airframe component for the aerial vehicle.

[0064] Referring to FIG. **3**, each of the vehicle airfoils **112** may be configured as a wing, a stabilizer and/or any other type of projection including and/or forming a control surface for the vehicle. The vehicle airfoils **112** of FIG. **3** are distributed circumferentially about the vehicle body

110 and the axial centerline **28**. Referring to FIGS. **4** and **5**, each of the vehicle airfoils **112** may be connected to the vehicle body **110**. Each of the vehicle airfoils **112** projects spanwise (e.g., radially relative to the axial centerline **28**) along a span line of the respective vehicle airfoil **112** out from the vehicle body **110** to a tip **116** of the respective vehicle airfoil **112**. Each vehicle airfoil **112** of FIGS. **4** and **6** extends chordwise along a chord line of the respective vehicle airfoil **112** between and to a leading edge **118** of the respective vehicle airfoil **112** and a trailing edge **120** of the respective vehicle airfoil **112**. Each vehicle airfoil **112** of FIGS. **5** and **6** has an exterior first surface **122** and an exterior second surface **124**, and extends laterally between and to its first surface **122** and its second surface **124**. The first surface **122** and the second surface **124** of FIG. **6** extend chordwise between and may meet at the respective leading edge **118** and the respective trailing edge **120**. The first surface **122** and the second surface **124** of FIG. **5** extend spanwise from the vehicle body **110** to (and may meet at) the respective airfoil tip **116**.

[0065] Referring to FIG. **1**, the fluid control system **26** includes a system intake **126** and one or more fluid circuits **128**, where each fluid circuit **128** may be (e.g., uniquely) associated with a respective one of the vehicle airfoils **112**. The system intake **126** may be configured as or otherwise includes an intake duct **130**; e.g., a bleed duct. Referring to FIG. **2**, the intake duct **130** is formed by and is radially between an inner duct wall **132** and an outer duct wall **134**. The inner duct wall **132** of FIG. **2** may be formed by/include at least the downstream section **74** of the compressor case **76** and at least a forward, upstream section of the diffuser case **86**. The outer duct wall **134** extends circumferentially about (e.g., circumscribes), axially overlaps and is spaced radially outward from the inner duct wall **132**. A forward, upstream section of the outer duct wall **134** may be formed by/included in the upstream stationary structure **64**. An aft, downstream section of the outer duct wall **134** may be formed by/included in the downstream stationary structure **66**.

[0066] The intake duct **130** has an (e.g., annular or arcuate) interior flowpath **136** between the inner duct wall **132** and the outer duct wall **134**. This duct flowpath **136** is fluidly coupled with the core flowpath **34** through an inlet **138** to the fluid control system **26**. Referring to FIG. **7A**, the system inlet **138** may be configured from a single inlet aperture **140** through the compressor case **76**. This inlet aperture **140** may be an annular slot which extends circumferentially (e.g., uninterrupted) completely around the axial centerline **28**. Alternatively, referring to FIG. **7B**, the system inlet **138** may be configured from one or more inlet apertures **140'** through the compressor case **76**. Each of these inlet apertures **140'** may be an arcuate slot which extends circumferentially (e.g., uninterrupted) partially about the axial centerline **28**.

[0067] Referring to FIG. **2**, the system inlet **138** may be disposed at an intermediate (e.g., axial midpoint) location along the compressor section **37** and its compressor rotor **44**. However, in other embodiments, the system inlet **138** may be disposed at an aft, downstream end of the compressor rotor **44**, or may be spaced aft, downstream of the compressor rotor **44** along the core flowpath **34**. In still other embodiments, the system inlet **138** may be disposed at a forward, upstream end of the compressor rotor **44**, or may be spaced forward, upstream of the compressor rotor **44** along the core flowpath **34**. The intake duct **130** and its duct flowpath **136** may thereby receive (e.g., bleed) the core air from the core flowpath **34** at various locations along the core flowpath **34**.

[0068] Referring to FIG. **8**, each fluid circuit **128** includes a manifold **142**, a flow regulator **144** (e.g., a valve) and one or more internal airfoil passages **146** and **148**. The manifold **142** is disposed radially between the gas turbine engine **22** and the vehicle body **110**. The manifold **142** of FIG. **8**, for example, projects radially out from the downstream stationary structure **66** and its outer duct wall downstream section to a sidewall **150** of the vehicle body **110** (see FIG. **1**). The manifold **142** may thereby be connected to (e.g., fixedly attached to) the respective vehicle airfoil **112** through the body sidewall **150**. Alternatively, the manifold **142** may extend through an aperture in the body sidewall **150** to a base **152** of the respective vehicle airfoil **112**. The manifold **142** may thereby be directly connected to (e.g., fixedly attached to) the respective vehicle airfoil **112**. Referring to FIG. **1**, each manifold **142** may be formed integral with the downstream stationary structure **66** and

included in the monolithic body **70**.

[0069] The manifold **142** of FIG. **8** includes an inlet passage **154** and one or more outlet passages **156** and **158**. The inlet passage **154** is configured to fluidly couple each of the outlet passages **156** and **158** to the intake duct **130** and its duct flowpath **136** through the flow regulator **144**. The first outlet passage **156** is configured to fluidly couple the first airfoil passage **146** to the inlet passage **154** through the flow regulator **144**. The second outlet passage **158** is configured to fluidly couple the second airfoil passage **148** to the inlet passage **154** through the flow regulator **144**. The outlet passages **156** and **158** split off from the inlet passage **154** in parallel.

[0070] The flow regulator **144** of FIG. **8** is arranged with (e.g., within) the manifold **142**. The flow regulator **144** may be configured as a two-way valve. The flow regulator **144**, for example, may fluidly couple the inlet passage **154** to the first outlet passage **156** during a first mode. The flow regulator **144** may also fluidly decouple the inlet passage **154** from the second outlet passage **158** during the first mode. The flow regulator **144** may fluidly couple the inlet passage **154** to the second outlet passage **158** during a second mode. The flow regulator **144** may also fluidly decouple the inlet passage **154** from the first outlet passage **156** during the second mode.

[0071] Referring to FIGS. **9A** and **9B**, the flow regulator **144** may include a regulator (e.g., valve) body **160** and a movable regulator (e.g., valve) element **162**. The regulator body **160** may be formed as an integral portion of the manifold **142**. The regulator element **162** may be configured as a pivotable flap. The regulator element **162**, for example, may be pivotable between an open first position (e.g., see FIG. **9A**) and an open second position (e.g., see FIG. **9B**). When the regulator element **162** is in the first position of FIG. **9A**, the regulator element **162** uncovers an inlet opening **164** to the first outlet passage **156** and covers an inlet opening **166** to the second outlet passage **158**. When the regulator element **162** is in the second position of FIG. **9B**, the regulator element **162** covers the inlet opening **164** to the first outlet passage **156** and uncovers the inlet opening **166** to the second outlet passage **158**. Of course, various other types of flow regulator configurations are known in the art, and the present disclosure is not limited to any particular ones thereof.

[0072] Referring to FIGS. **5** and **6**, each of the airfoil passages **146** and **148** extends within the respective vehicle airfoil **112**. Each airfoil passage **146**, **148** of FIG. **5**, for example, extends spanwise from the airfoil base **152** towards (e.g., to) an endwall of the respective vehicle airfoil **112** forming the airfoil tip **116**. Referring to FIG. **6**, each airfoil passage **146**, **148** may be disposed at (e.g., on, adjacent or proximate) the airfoil leading edge **118**. Referring to FIGS. **5** and **6**, each airfoil passage **146**, **148** may be fluidly coupled with one or more respective apertures **168**, **170** in a respective one of the airfoil surfaces **122**, **124**. The first apertures **168** of FIG. **5** are distributed spanwise along the first airfoil passage **146**, and project into the respective vehicle airfoil **112** from the airfoil first surface **122**. The second apertures **170** of FIG. **5** are distributed spanwise along the second airfoil passage **148**, and project into the respective vehicle airfoil **112** from the airfoil second surface **124**. Note, while the airfoil apertures **168** and **170** are shown in linear arrays at the leading edge **118**, the present disclosure is not limited to such an exemplary arrangement. The airfoil apertures **168**, **170**, for example, may be arranged in multiple arrays and/or may be disposed at an intermediate or other chordwise location along the respective airfoil surface **122**, **124**.

Furthermore, while the airfoil apertures **168** and **170** are shown with circular geometries, one or more or all of the airfoil apertures **168** and/or **170** may each alternatively have a polygonal and/or elongated geometry. For example, one or more of the airfoil apertures **168**, **170** may be replaced with an elongated slot.

[0073] Referring to FIG. **2**, during gas turbine engine operation, a (e.g., relatively small) quantity of the core gas may be directed (e.g., bled) from the core flowpath **34** into the intake duct **130** and its duct flowpath **136** through the system inlet **138**. This bled core gas is directed into one or more or all of the fluid circuits **128** through the inlet passage(s) **154** as control gas.

[0074] Within the fluid circuit **128** of FIG. **8** and during the first mode of FIG. **9A**, (e.g., all of) the control gas within the inlet passage **154** is directed through the flow regulator **144** to the first outlet

passage **156**. The control gas subsequently flows into and through the first airfoil passage **146**. The first apertures **168** direct (e.g., exhaust) this control gas out of the respective vehicle airfoil **112** into an external environment **172** to flow along the airfoil first surface **122**; see FIG. **6**. Referring to FIG. **10A**, the exhausted control gas flowing into the external environment **172** and/or along the airfoil first surface **122** may functionally (e.g., virtually) change an aerodynamic characteristic of the respective vehicle airfoil **112**. More particularly, the exhaust control gas may form a layer of boundary layer gas **174** between the airfoil first surface **122** and ambient air flowing along the airfoil first surface **122** thereby functionally displacing the airfoil first surface **122** laterally outward. The exhaust control gas may therefore functionally change an aerodynamic profile of the respective vehicle airfoil **112**.

[0075] By contrast, within the fluid circuit **128** of FIG. **8** and during the second mode of FIG. **9B**, (e.g., all of) the control gas within the inlet passage **154** is directed through the flow regulator **144** to the second outlet passage **158**. The control gas subsequently flows into and through the second airfoil passage **148**. The second apertures **170** direct (e.g., exhaust) this control gas out of the respective vehicle airfoil **112** into the external environment **172** to flow along the airfoil second surface **124**; see FIG. **6**. Referring to FIG. **10B**, the exhausted control gas flowing into the external environment **172** and/or along the airfoil second surface **124** may functionally (e.g., virtually) change an aerodynamic characteristic of the respective vehicle airfoil **112**. More particularly, the exhaust control gas may form a layer of boundary layer gas **174** between the airfoil second surface **124** and ambient air flowing along the airfoil second surface **124** thereby functionally displacing the airfoil second surface **124** laterally outward. The exhaust control gas may therefore functionally change the aerodynamic profile of the respective vehicle airfoil **112**.

[0076] By selectively directing the control gas out of the first apertures **168** and/or the second apertures **170**, the fluid control system **26** may configure the airfoil first surface **122** as a pressure side surface and the airfoil second surface **124** as a suction side surface, or vice versa. The fluid control system **26** may thereby maneuver (e.g., control a flight direction/a flight path of) the aerial vehicle by selectively exhausting the control gas out of the first apertures **168** and/or the second apertures **170**. By using the control gas to influence/control aerial vehicle flight, the vehicle airfoils **112** may be configured as fixed/stationary airfoils. The aerial vehicle may therefore be configured without traditional (e.g., electric, hydraulic, pneumatic, etc.) actuators for controlling (e.g., trimming) variable position airfoils. This may reduce the complexity, weight, size and cost of the aerial vehicle.

[0077] The flow regulator **144** is described above with respect to FIGS. **9A** and **9B** as being in open-closed positions; e.g., a position where one of the outlet passages **156**, **158** is opened and where one of the outlet passages **158**, **156** is closed. However, in some embodiments, the flow regulator **144** may also be configured to open both of the outlet passages **156** and **158**. For example, referring to FIG. **11**, the regulator element **162** may be pivoted to an intermediate position between the first position of FIG. **9A** and the second position of FIG. **9B** that uncovers both passage inlets **164** and **166**.

[0078] In some embodiments, referring to FIGS. **9A**, **9B** and **11**, each flow regulator **144** may always direct at least some or all of the control gas into one or more of the outlet passages **156** and **158**. In other embodiments however, referring to FIGS. **12A-C** or FIGS. **13A-C**, the flow regulator **144** may alternatively be configured to shut off control gas flow or direct the control gas elsewhere during at least one mode of operation.

[0079] Referring to FIGS. **12A-C**, the regulator element **162** may include one or more side channels **176** and **178**; e.g., grooves. The first side channel **176** is disposed on a first side **180** of the regulator element **162**. The second side channel **178** is disposed on a second side **182** of the regulator element **162**. The first side channel **176** is configured to provide a fluid coupling between the inlet passage **154** and the first outlet passage **156** when the regulator element **162** is in its first position of FIG. **12A**. However, when the regulator element **162** is in the intermediate position of

FIG. 12B or the second position of FIG. 12C, an end **184** of the first side channel **176** fluidly decouples the inlet passage **154** from the first outlet passage **156**. Similarly, the second side channel **178** is configured to provide a fluid coupling between the inlet passage **154** and the second outlet passage **158** when the regulator element **162** is in its second position of FIG. 12C. However, when the regulator element **162** is in the intermediate position of FIG. 12B or the first position of FIG. 12A, an end **186** of the second side channel **178** fluidly decouples the inlet passage **154** from the second outlet passage **158**. With this arrangement, the regulator element **162** may be positioned in its intermediate position of FIG. 12B to cutoff the flow of the control gas to both of the outlet passages **156** and **158** and, thus, the flow of the control gas to both of the airfoil passages **146** and **148**; see FIG. 6.

[0080] The regulator element **162** may have a tapered (e.g., teardrop-shaped) cross-sectional geometry when viewed, for example, in a reference plane perpendicular to a pivot axis **188** of the regulator element **162**. The regulator element **162** of FIGS. 12A-C, for example, includes an arcuate base surface **190**, a first side surface **192** and a second side surface **194**. The arcuate base surface **190** extends circumferentially about the pivot axis **188** between and to downstream ends of the side surfaces **192** and **194**. The first side surface **192** is disposed at the first side **180** of the regulator element **162**, and may have a planar (e.g., straight line) geometry. The second side surface **194** is disposed at the second side **182** of the regulator element **162**, and may have a planar (e.g., straight line) geometry. These side surfaces **192** and **194** extend along the element sides **180** and **182** to and may meet at a leading edge **196** of the regulator element **162**. The first side channel **176** projects laterally into the regulator element **162** from the first side surface **192**. This first side channel **176** projects longitudinally into the regulator element **162** from the leading edge **196** to the respective channel end **184**. The second side channel **178** projects laterally into the regulator element **162** from the second side surface **194**. This second side channel **178** projects longitudinally into the regulator element **162** from the leading edge **196** to the respective channel end **186**.

[0081] Referring to FIGS. 13A-C, the regulator element **162** may have a similar configuration to that described with respect to FIGS. 12A-C. However, the regulator element **162** of FIGS. 13A-C also includes at least one internal volume **198**; e.g., a passage, a bore, etc. This internal volume **198** is configured to align with an aperture **200** in a pin **202** (e.g., a pivot shaft) for the regulator element **162** in its intermediate position of FIG. 13B. The internal volume **198** may thereby fluidly couple the inlet passage **154** to a third outlet passage **204** (e.g., a bore) extending within/through the pin **202** when the regulator element **162** is in the intermediate position. However, when the regulator element **162** is in the first position of FIG. 13A or the second position of FIG. 13C, the internal volume **198** may be (e.g., completely) misaligned with the aperture **200** thereby fluidly decoupling the inlet passage **154** from the third outlet passage **204**. A sidewall of the pin **202**, for example, may block/cover an outlet from the internal volume **198** when the regulator element **162** is in the first position of FIG. 13A or the second position of FIG. 13C. The internal volume **198** of FIGS. 13A-C may project longitudinally into the regulator element **162** from the leading edge **196** to its outlet within a central region of the regulator element **162**.

[0082] Referring to FIG. 14, the third outlet passage **204** may be fluidly coupled with a plenum **206** (see also FIG. 1) within the vehicle airframe **24**. This plenum **206** may be between and formed by the gas turbine engine **22** and its static engine structure **42** and the vehicle body **110** and its sidewall **150**. The fluid circuit **128** may thereby vent at least some or all of the control gas when the control gas is not needed for controlling aerial vehicle flight. Within the plenum **206**, the vented control gas may also provide gas turbine engine cooling. In other embodiments, however, the third outlet passage **204** may also or alternatively vent the control gas into the exterior environment **172**. Alternatively, referring to FIG. 15, the third outlet passage **204** may be fluidly coupled with the core flowpath **34** at a point downstream of the system inlet **138**. The fluid circuit **128** may thereby return at least some or all of the control gas to the gas turbine engine **22** for operation when the control gas is not needed for controlling aerial vehicle flight.

[0083] The gas turbine engine 22 is described above as a single spool, radial-flow turbojet gas turbine engine for ease of description. The present disclosure, however, is not limited to such an exemplary gas turbine engine. The gas turbine engine 22, for example, may alternatively be configured as an axial flow gas turbine engine. The gas turbine engine 22 may be configured as a direct drive gas turbine engine. The gas turbine engine 22 may alternatively include a gear train that connects one or more rotors together such that the rotors rotate at different speeds. The gas turbine engine 22 may be configured with a single spool (e.g., see FIG. 1), two spools, or with more than two spools. The gas turbine engine 22 may be configured as a turbofan engine, a turbojet engine, a propfan engine, a pusher fan engine or any other type of turbine engine. In addition, while the gas turbine engine 22 is described above with an exemplary reverse flow annular combustor, the gas turbine engine 22 may also or alternatively include any other type/configuration of annular, tubular (e.g., CAN), axial flow and/or reverse flow combustor. The present disclosure therefore is not limited to any particular types or configurations of turbine engines.

[0084] While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly, comprising: a component outside of a gas turbine engine; and a fluid circuit configured to bleed gas from a core flowpath of the gas turbine engine to provide bleed gas, the fluid circuit configured to direct the bleed gas to the component during a first mode, and the fluid circuit configured to direct the bleed gas back to the core flowpath or into a plenum outside of the gas turbine engine during a second mode.
2. The assembly of claim 1, wherein the fluid circuit is configured to direct the bleed gas into the plenum outside of the gas turbine engine during the second mode.
3. The assembly of claim 1, wherein the fluid circuit is configured to direct the bleed gas back to the core flowpath during the second mode.
4. The assembly of claim 1, wherein the component comprises an airfoil configured as a control surface.
5. The assembly of claim 1, wherein the fluid circuit is configured to direct the bleed gas to a first portion of the component during the first mode; and the fluid circuit is configured to direct the bleed gas to a second portion of the component during a third mode.
6. The assembly of claim 1, wherein the fluid circuit includes a movable regulator element; the fluid circuit is configured to flow the bleed gas to a side of the movable regulator element during the first mode; and the fluid circuit is configured to flow the bleed gas through an internal volume of the movable regulator element during the second mode.
7. An assembly, comprising: a regulator manifold including an inlet passage, a first outlet passage and a second outlet passage, the inlet passage configured to receive bleed gas from a core flowpath of a gas turbine engine; and a regulator element within the regulator manifold and configured to pivot to a first position, a second position and an intermediate position between the first position and the second position, the regulator element configured to fluidly couple the inlet passage with the first outlet passage in the first position, the regulator element configured to fluidly couple the inlet passage with the second outlet passage in the second position, and the regulator element configured to fluidly decouple the inlet passage from the first outlet passage and the second outlet

passage in the intermediate position; wherein the regulator manifold further includes a third outlet passage, the regulator element is configured to fluidly couple the inlet passage to the third outlet passage in the intermediate position, and the third outlet passage is configured to fluidly couple the inlet passage with a plenum outside of the gas turbine engine when the regulator element is in the intermediate position.

8. The assembly of claim 7, wherein the regulator element is configured to fluidly decouple the inlet passage from the second outlet passage in the first position; and the regulator element is configured to fluidly decouple the inlet passage from the first outlet passage in the second position.

9. The assembly of claim 7, further comprising: an airfoil extending laterally between a first surface and a second surface, the airfoil including a first aperture in the first surface and a second aperture in the second surface; the first outlet passage fluidly coupled with the first aperture; and the second outlet passage fluidly coupled with the second aperture.

10. The assembly of claim 9, further comprising: an airframe including a body and the airfoil projecting out from the body; and the gas turbine engine mounted to the airframe.

11. The assembly of claim 7, further comprising: an airfoil comprising a first surface and a first aperture in the first surface; and the first outlet passage fluidly coupled with the first aperture.

12. The assembly of claim 11, wherein the airfoil further comprises a second surface and a second aperture in the second surface; and the second outlet passage is fluidly coupled with the second aperture.

13. The assembly of claim 7, wherein the regulator element includes a first channel and a second channel; the first channel fluidly couples the inlet passage with the first outlet passage when the regulator element is in the first position; and the second channel fluidly couples the inlet passage with the second outlet passage when the regulator element is in the second position.

14. The assembly of claim 7, wherein the gas turbine engine comprises a rotating structure and a stationary structure; the rotating structure includes a compressor rotor within a compressor section of the gas turbine engine and a turbine rotor within a turbine section of the gas turbine engine; the stationary structure at least partially houses the rotating structure; and the regulator manifold is formed integral with the stationary structure.

15. The assembly of claim 14, wherein the stationary structure houses the turbine rotor.

16. The assembly of claim 14, wherein the stationary structure comprises a combustor within a combustor section of the gas turbine engine.
